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EDUCATION

Shandong University, Qingdao
2024 – 2028

Institute of Frontier and Interdisciplinary Science · class of 2024, expected 2028

Coursework: theoretical mechanics, electrodynamics, thermodynamics and statistical physics, quantum mechanics, advanced quantum mechanics, quantum field theory, particle physics, general relativity. The second half of that list is where the current work sits.

Reading and reproduction — light-front and TMD
2024 – now

Working through the light-front quantisation and TMD literature paper by paper, re-deriving the key relations symbolically before trusting them, and keeping the checks in a repository so each claim can be re-run.

HOW I WORK

Derive from first principles, not from precedent — ugly is fine, fabricated is not. Every claim has to replay from a log.

RESEARCH INTERESTS

I am interested in how much of hadron structure can be derived rather than fitted.

Light-front QCD. Quantise on the light front and the vacuum turns trivial — the difficulty does not disappear, it reappears in the Fock truncation and in the zero modes the $A^+=0$ gauge leaves behind. Next for me: the holographic link from confinement scale to bound-state equation.

Hadron structure from Fock wave functions. A transverse-momentum distribution read as a spectator-integrated amplitude, not a fitting function, with $\Sigma x_i=1$ and $\Sigma \mathbf{k}_{\perp i}=0$ built in from the start.

TMD factorisation. Collins–Soper evolution and its non-perturbative kernel; the SIDIS structure-function dictionary (Sivers, Collins, Boer–Mulders); the Drell–Yan sign reversal that keeps SIDIS and DY from being fitted apart.

RESEARCH DIRECTIONS & WORK IN PROGRESS

- **Light-front quantisation (PH 01).** Working toward: an answer to where the Fock-space truncation can still be trusted. The two handles I am following are DLCQ matrices whose UV cutoff is matched to the truncation, and what the zero modes keep carrying once the gauge is fixed.
- **Baryon light-cone Fock expansion (PH 02).** Working toward: a truncation checkable against constraints that do not bend — normalisation ($\Sigma_n \int |\psi_n|^2=1$, $\int dx\, f_1$ equal to the valence count) and the mass relation $P^+P^- - P_\perp{}^2 = M^2$ — three-quark sector first, then five, and which observables survive a cut at fixed N.
- **Proton structure — DIS / SIDIS / DY (PH 03).** Working toward: one picture the three processes agree on. TMD factorisation at $P_{hT} \ll Q$, matched onto a Y-term at large transverse momentum, Collins–Soper evolution with a non-perturbative kernel, and where a Gaussian width in $\langle k_\perp{}^2 \rangle$ gives way to power-law tails.

CURRENT WORK

- **PH 02 · Baryon light-front wave function at fixed N.** three-quark sector first, then how far the five-quark component moves the structure function.
- **PH 03 · Cross-process constraints.** DIS, SIDIS and DY in one parameterisation, starting where they disagree.

OPEN QUESTIONS

- **PH 01 · Light-front quantization.** Does the multiplicative kinematics actually save work, or does it just move the work into the Fock space? · If the vacuum got simpler, which physics got moved into the wave function instead? · Which observable is the right monitor for truncation error — and how would I know, before I have the exact answer?
- **PH 02 · Baryon light-cone Fock expansion.** Is convergence driven by cutting $k\perp$ or by adding Fock components? · Which observables are least sensitive to the cut — and are they the interesting ones? · In which process do five-quark components rise above the truncation error?
- **PH 03 · Proton structure — DIS / SIDIS / DY.** Do SIDIS and DY disagree on the same $\langle k\perp^2 \rangle$ — and is a disagreement a sign reversal or an evolution effect? · In which Q^2 range does the Gaussian width model start to fail? · Which term of the 3D distribution is shortest on data — and is that a fact about the data, or about the factorisation?

TECHNICAL SKILLS

LANGUAGES	Python for numerics and fits, C++ for compute, Bash, Git
SYMBOLIC	Mathematica for algebra; SymPy — mathkit, an open-source skill I maintain; SymPy for Agent (upcoming); numerical integration and fitting; ROOT for plots and data; convolution and evolution codes
THEORY	Light-front quantisation, Fock-basis expansion, TMD factorisation and evolution, PDF and TMD fits, perturbative calculations
TOOLING	LaTeX and TikZ, reproducible notes, Linux and HPC, papers and data management

LANGUAGES

Chinese (native) · English (working proficiency).